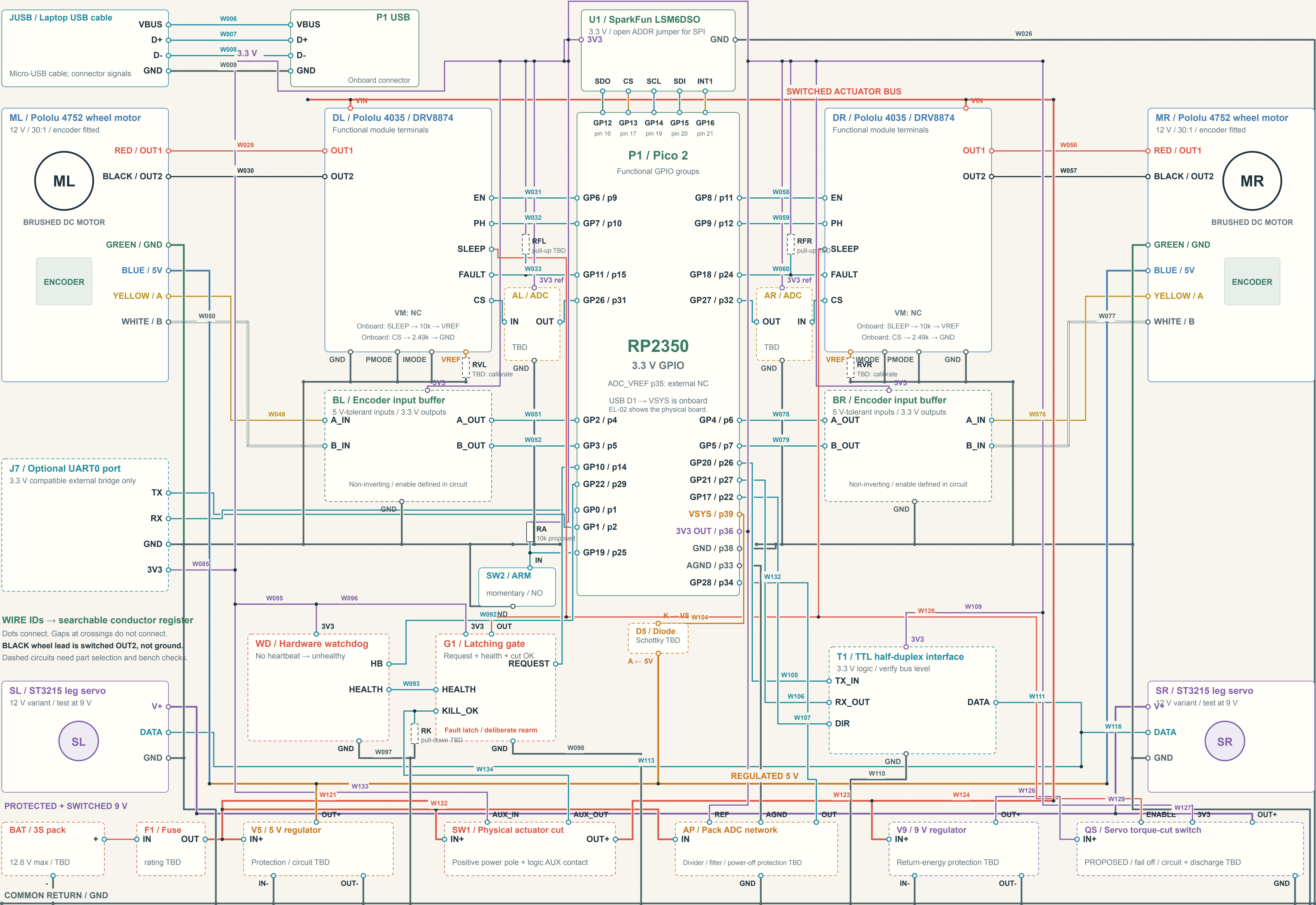


Complete system wiring

Every external conductor is drawn. Both wheel motors, both encoders and both leg servos have separate terminals.



Pico 2 + IMU connections

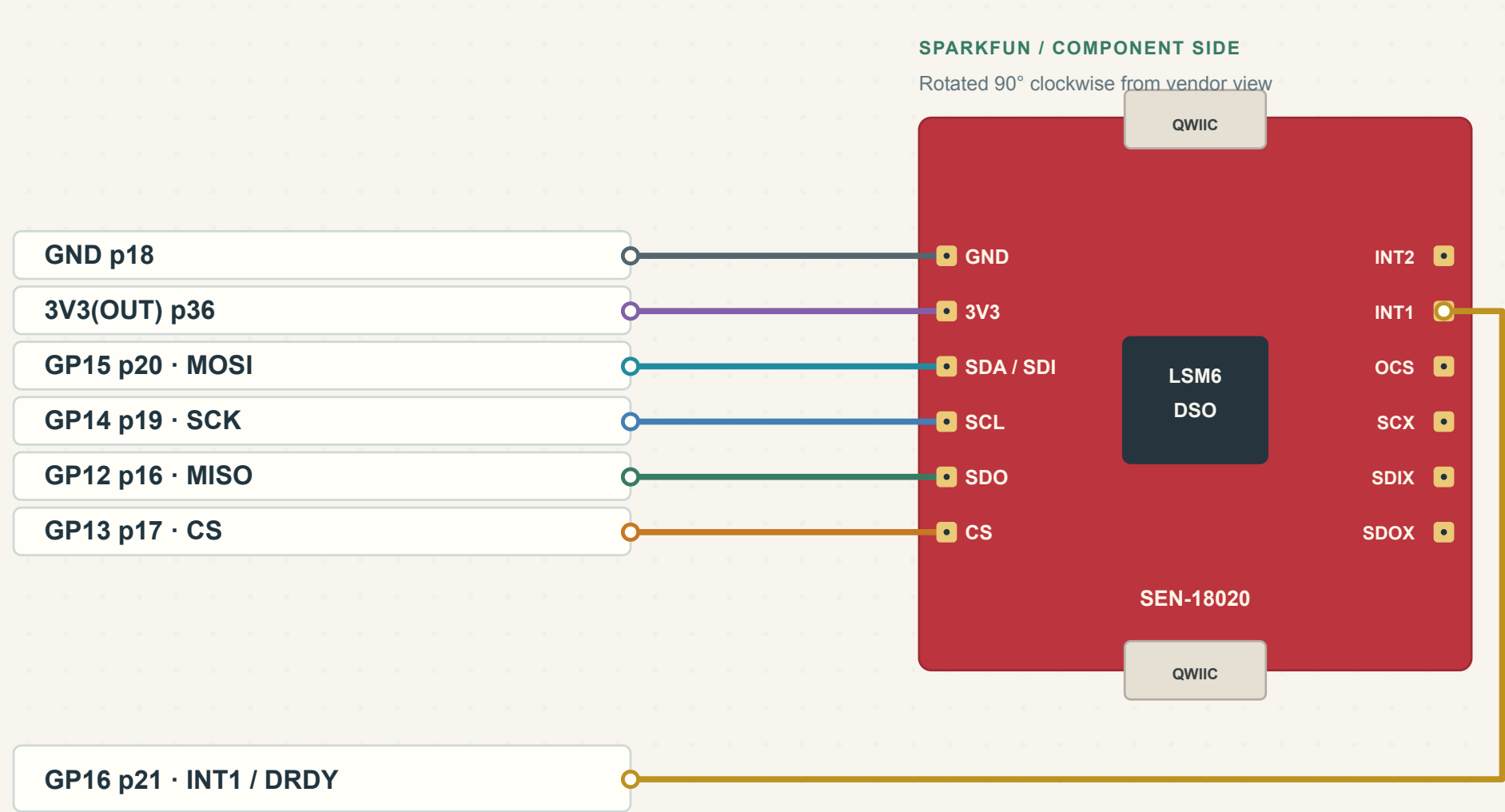
Component-side views, USB at the top of the Pico. Physical pins and GPIO numbers are distinct.

SOURCE-CHECKED PLAN
HARNESS / BENCH RELEASE OPEN

PHYSICAL PIN MAP

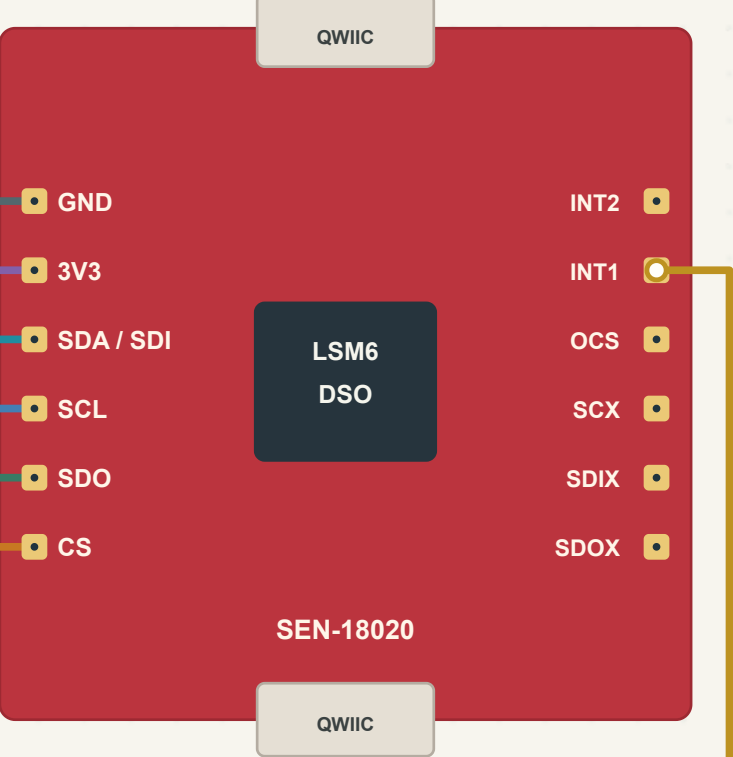


SEVEN-WIRE SENSOR HARNESS



SPARKFUN / COMPONENT SIDE

Rotated 90° clockwise from vendor view



SPI JUMPER SETUP / CHECK RECEIVED BOARD

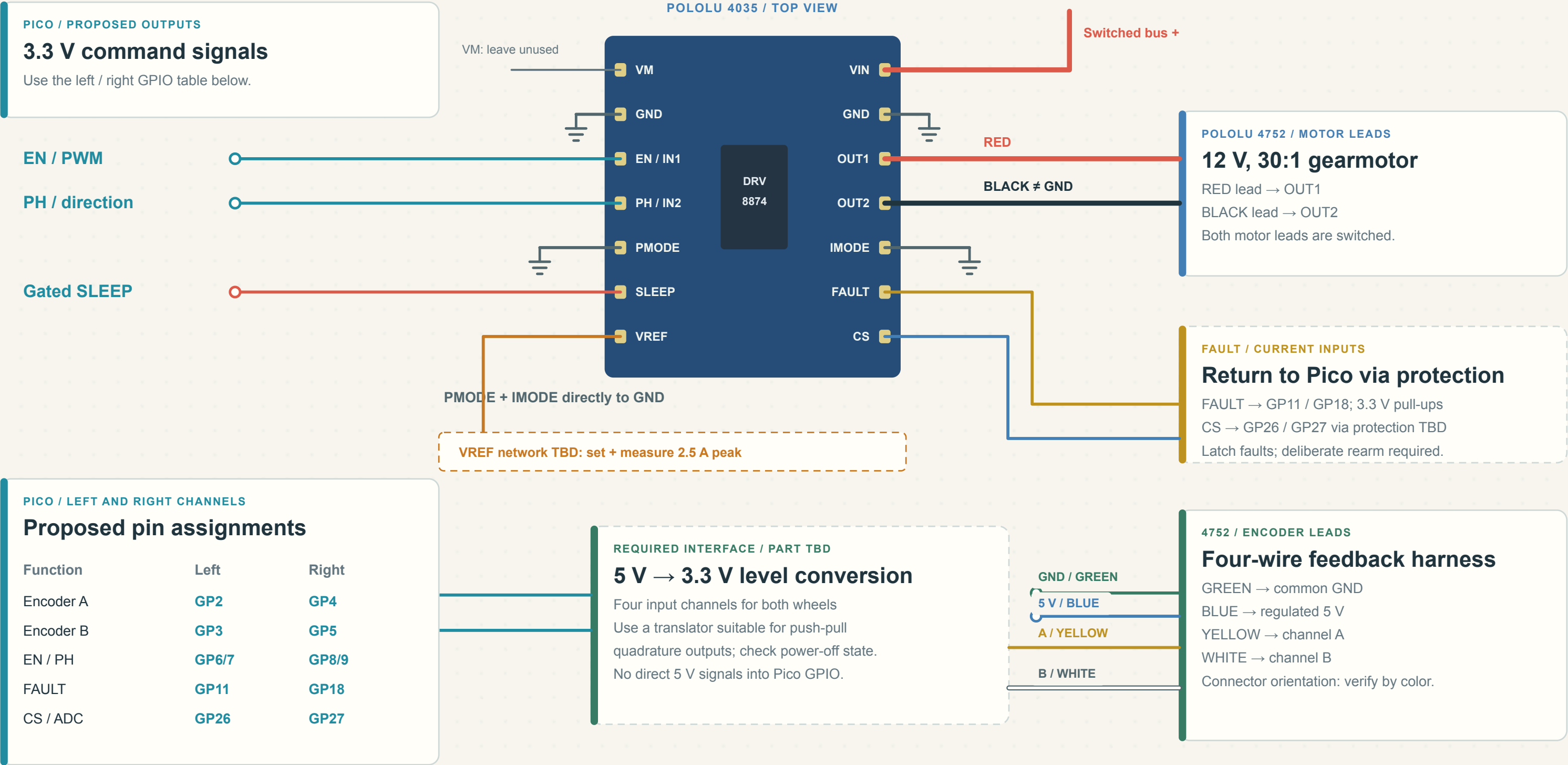
Open the 0x6B / 0x6A address jumper so SDO can drive MISO.
Recommended: open both I2C pull-up traces. Keep SCX / SDIX ground jumpers intact.
Qwiic sockets carry I2C and are unused here. Wire the main SPI pads, not SCX / SDIX.
Power and signal logic are 3.3 V. Inspect board revision and check sensor identity before use.

p1 is upper left; p40 is upper right.

Wheel motor + encoder harness

One channel shown; repeat for the other wheel using the assignment table. Colors match the motor leads.

SOURCE-CHECKED PLAN
HARNESS / BENCH RELEASE OPEN



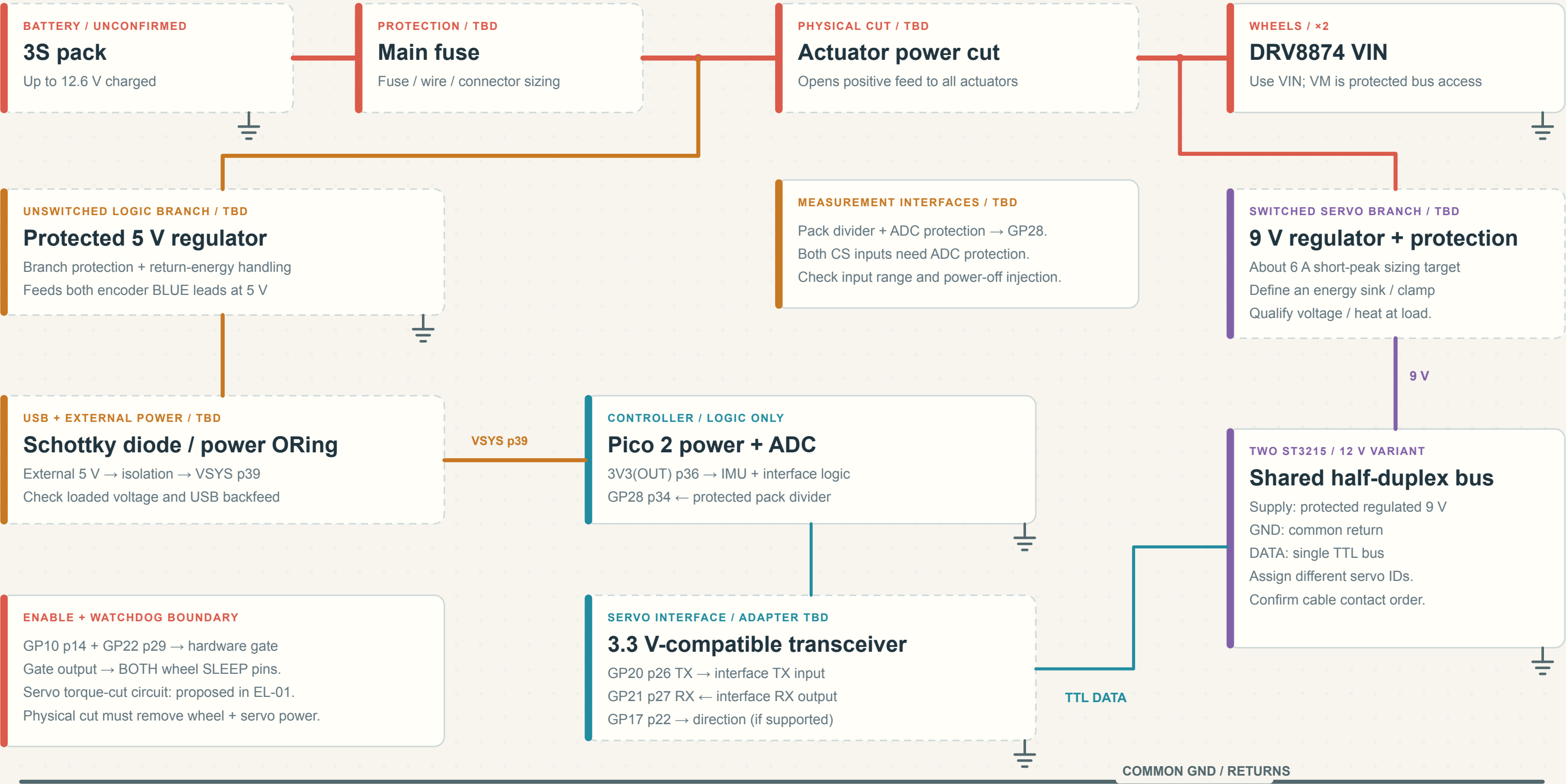
EN low = brake. SLEEP low = outputs off / coast.

2.5 A peak / 1.2 A RMS are Hux qualification targets, not tested settings.

Power distribution + servo bus

Functional interface drawing: regulator, clamp, fuse, enable-gate and bus-adapter circuits remain to select.

SOURCE-CHECKED PLAN
HARNESS / BENCH RELEASE OPEN



All GND symbols join this reference. Route high-current returns away from sensor / MCU wiring.

Both wheel motors — every wire

Two complete channels. Rails and enable come from EL-01. Use terminal names here. EL-01 has the conductor register IDs.

SOURCE-CHECKED PLAN
HARNESS / BENCH RELEASE OPEN

